

ACCURATE AND SCALABLE ESTIMATION OF EPISTEMIC UNCERTAINTY FOR GRAPH NEURAL NETWORKS

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ABSTRACT

While graph neural networks (GNNs) are widely used for node and graph representation learning tasks, the reliability of GNN uncertainty estimates under distribution shifts remains relatively under-explored. Indeed, while post-hoc calibration strategies can be used to improve in-distribution calibration, they need not also improve calibration under distribution shift. However, techniques which produce GNNs with better intrinsic uncertainty estimates are particularly valuable, as they can always be combined with post-hoc strategies later. Therefore, in this work, we propose G- Δ UQ, a novel training framework designed to improve intrinsic GNN uncertainty estimates. Our framework adapts the principle of stochastic data centering to graph data through novel graph anchoring strategies, and is able to support partially stochastic GNNs. While, the prevalent wisdom is that fully stochastic networks are necessary to obtain reliable estimates, we find that the functional diversity induced by our anchoring strategies when sampling hypotheses renders this unnecessary and allows us to support G- Δ UQ on pretrained models. Indeed, through extensive evaluation under covariate, concept and graph size shifts, we show that G- Δ UQ leads to better calibrated GNNs for node and graph classification. Further, it also improves performance on the uncertainty-based tasks of out-of-distribution detection and generalization gap estimation. Overall, our work provides insights into uncertainty estimation for GNNs, and demonstrates the utility of G- Δ UQ in obtaining reliable estimates.

1 INTRODUCTION

As graph neural networks (GNNs) are increasingly deployed in critical applications with test-time distribution shifts (Zhang & Chen, 2018; Gaudelet et al., 2020; Yang et al., 2018; Yan et al., 2019; Zhu et al., 2022), it becomes necessary to expand model evaluation to include safety-centric metrics, such as calibration errors (Guo et al., 2017), out-of-distribution (OOD) rejection rates (Hendrycks & Gimpel, 2017), and generalization error predictions (GEP) (Jiang et al., 2019), to holistically understand model performance in such shifted regimes (Hendrycks et al., 2022b; Trivedi et al., 2023b). Notably, improving on these additional metrics often requires reliable uncertainty estimates, such as maximum softmax or predictive entropy, which can be derived from prediction probabilities. Although there is a clear understanding in the computer vision literature that the quality of uncertainty estimates can noticeably deteriorate under distribution shifts (Wiles et al., 2022; Ovadia et al., 2019), the impact of such shifts on graph neural networks (GNNs) remains relatively under-explored.

Post-hoc calibration methods (Guo et al., 2017; Gupta et al., 2021; Kull et al., 2019; Zhang et al., 2020), which use validation datasets to rescale logits to obtain better calibrated models, are an effective, accuracy-preserving strategy for improving uncertainty estimates and model trust-worthiness. Indeed, several post-hoc calibration strategies (Hsu et al., 2022; Wang et al., 2021) have been recently proposed to explicitly account for the non-IID nature of node-classification datasets. However, while these methods are effective at improving uncertainty estimate reliability on in-distribution (ID) data, they have not been evaluated on OOD data, where they may become unreliable. To this end, training strategies which produce models with better intrinsic uncertainty estimates are valuable as they will provide better out-of-the-box ID and OOD estimates, which can then be further combined with post-hoc calibration strategies if desired.

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The Δ -UQ training framework (Thiagarajan et al., 2022) was recently proposed as a scalable, single model alternative for vision models ensembles and has achieved state-of-the-art performance on calibration and OOD detection tasks. Central to Δ -UQ’s success is the concept of *anchored* training, where models are trained on stochastic, relative representations of input samples in order to simulate sampling from different functional modes at test time (Sec. 2.) While, on the surface, Δ -UQ also appears as a potentially attractive framework for obtaining reliable, intrinsic uncertainty estimates on graph-based tasks, there are several challenges that arise from the structured, discrete, and variable-sized nature of graph data that must be resolved first. Namely, the anchoring procedure used by Δ -UQ is not applicable for graph datasets, and it is unclear how to design alternative anchoring strategies such that sufficiently diverse functional modes are sampled at inference to provide reliable epistemic uncertainty estimates.

Proposed Work. Thus, our work proposes G- Δ UQ, a novel training paradigm which provides better intrinsic uncertainty estimates for both graph and node classification tasks through the use of newly introduced graph-specific, anchoring strategies. Our contributions can be summarized as follows:

- **(Partially) Stochastic Anchoring for GNNs.** We propose G- Δ UQ, a novel training paradigm that improves the reliability of uncertainty estimates on GNN-based tasks. Our novel graph-anchoring strategies support partial stochasticity GNNs as well as training with pretrained models. (Sec. 3).
- **Evaluating Uncertainty-Modulated CIs under Distribution Shifts.** Across covariate, concept and graph-size shifts, we demonstrate that G- Δ UQ leads to better calibration. Moreover, G- Δ UQ’s performance is further improved when combined with post-hoc calibration strategies on several node and graph-level tasks, including new safety-critical tasks (Sec. 5).
- **Fine-Grained Analysis of G- Δ UQ.** We study the calibration of architectures of varying expressivity and G- Δ UQ’s ability to improve them under varying distribution shift. We further demonstrate its utility as a lightweight strategy for improving the calibration of pretrained GNNs (Sec. 6).

2 RELATED WORK & BACKGROUND

While uncertainty estimates are useful for a variety of safety-critical tasks (Hendrycks & Gimpel, 2017; Jiang et al., 2019; Guo et al., 2017), DNNs are well-known to provide poor uncertainty estimates directly out of the box (Guo et al., 2017). To this end, there has been considerable interest in building calibrated models, where the confidence of a prediction matches the probability of the prediction being correct. Notably, since GEP and OOD detection methods often rely upon transformations of a model’s logits, improving calibration can in turn improve performance on these tasks as well. Due to their accuracy-preserving properties, post-hoc calibration strategies, which rescale confidences after training using a validation dataset, are particularly popular. Indeed, several methods (Guo et al., 2017; Gupta et al., 2021; Kull et al., 2019; Zhang et al., 2020) have been proposed for DNNs in general and, more recently, dedicated node-classifier calibration methods (Hsu et al., 2022; Wang et al., 2021) have also been proposed to accommodate the non-IID nature of graph data. (See App. A.9 for more details.) Notably, however, such post-hoc methods do not lead to reliable estimates under distribution shifts, as enforcing calibration on ID validation data does not directly lead to reliable estimates on OOD data (Ovadia et al., 2019; Wiles et al., 2022; Hendrycks et al., 2019).

Alternatively, Bayesian methods have been proposed for DNNs (Hernández-Lobato & Adams, 2015; Blundell et al., 2015), and more recently GNNs (Zhang et al., 2019; Hasanzadeh et al., 2020), as inherently “uncertainty-aware” strategies. However, not only do such methods often lead to performance loss, require complicated architectures and additional training time, they often struggle to outperform the simple Deep Ensembles (DEns) baseline (Lakshminarayanan et al., 2017). By training a collection of independent models, DEns is able to sample different functional modes of the hypothesis space, and thus, capture epistemic variability to perform uncertainty quantification (Wilson & Izmailov, 2020). Given that DEns requires training and storing multiple models, the SoTA Δ -UQ framework (Thiagarajan et al., 2022) was recently proposed to sample different functional modes using only a single model, based on the principle of *anchoring*.

Background on Anchoring. Conceptually, anchoring is the process of creating a relative representation for an input sample in terms of a random “anchor.” By randomizing anchors throughout training (e.g., stochastically centering samples with respect to different anchors), Δ -UQ emulates the process of sampling and learning different solutions from the hypothesis space.

In detail, let $\mathcal{D}_{train}$ be the training distribution, $\mathcal{D}_{test}$ be the testing distribution, and $\mathcal{D}_{anchor} := \mathcal{D}_{train}$ be the anchoring distribution. Existing research on stochastic centering has focused on vision models (CNNs, ResNets, ViT) and used input space transformations to construct anchored representations. Specifically, given an image sample with corresponding label, $(\mathbf{I}, y)$, and anchor $\mathbf{C} \in \mathcal{D}_{anchor}$, anchored samples were created by subtracting and then channel-wise concatenating two images: $[\mathbf{I} - \mathbf{C} || \mathbf{C}]$ ¹. Given the anchored representation, a corresponding stochastically centered model can be defined as $f_\theta : [\mathbf{I} - \mathbf{C} || \mathbf{C}] \rightarrow \hat{y}$, and can be trained as shown in Fig. 1. At inference, similar to ensembles, predictions and uncertainties are aggregated over different hypotheses. Namely, given K random anchors, the mean target class prediction, $\mu(y|\mathbf{I})$, and the corresponding variance, $\sigma(y|\mathbf{I})$ are computed as: $\mu(y|\mathbf{I}) = \frac{1}{K} \sum_{k=1}^K f_\theta([\mathbf{I} - \mathbf{C}_k, \mathbf{C}_k])$ and $\sigma(y|\mathbf{I}) = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (f_\theta([\mathbf{I} - \mathbf{C}_k, \mathbf{C}_k]) - \mu)^2}$. Since the variance over K anchors captures epistemic uncertainty by sampling different hypotheses, these estimates can be used to modulate the predictions: $\mu_{calib} = \mu(1 - \sigma)$. Notably, the rescaled logits and uncertainty estimates have led to state-of-the-art performance on image outlier rejection, calibration, and extrapolation (Anirudh & Thiagarajan, 2022; Netanyahu et al., 2023).

3 GRAPH- Δ UQ: UNCERTAINTY-AWARE PREDICTIONS

Given Δ -UQ’s success in improving calibration and generalization (Netanyahu et al., 2023) under distribution shifts on computer vision tasks and the limitations of existing post-hoc strategies, stochastic centering appears as a potentially attractive framework for obtaining reliable uncertainty estimates when performing GNN-based classification tasks. However, there are several challenges that must be addressed before to applying it to graph data. Namely, while input space transformations, which induce fully stochastic models, were sufficient for sampling diverse functional hypotheses from vision models, it is (i) non-trivial to define such transformations when working with variable sized, discrete graph data and (ii) unclear whether full stochasticity is in fact needed when working with message passing models. Below, we explore these issues through novel graph anchoring strategies. However, we begin with a conceptual discussion of the role of anchoring strategies in generating reliable uncertainty estimates.

```
#training
for (I,y) in trainloader:
    C = create_anchors(n=BatchSize)
    I_Anc = CONCAT([I-C,C],dim=1)
    preds = model(I_Anc)
    loss = criterion(preds,Y)

#inference
anc_preds = []
testAnc = create_anchors(n=K)
for A in testAnc:
    I_anc = CONCAT([I-A,A],dim=1)
    preds = model(I_anc)
    anc_preds.append(preds)
P = CONCAT(anc_preds,dim=0)
mu = MEAN(P,dim=0)
var = STDDEV(P,dim=0)
```

Figure 1: **Training/Inference with Anchoring.**

What are the goals of anchoring?: As discussed in Sec. 2, epistemic uncertainty can be estimated by aggregating the variability over different functional hypotheses (Hüllermeier & Waegeman, 2021). Indeed, the prevalent wisdom behind the success of DeepEnS is its ability to sample *diverse* functional hypotheses. Since these hypotheses are more likely to differ on OOD inputs, aggregating them can lead to better generalization and uncertainty estimates. Insofar as stochastic centering seeks to simulate an ensemble through a single model, a key goal of the anchoring distribution/strategy is then to ensure that sampled hypotheses are also diverse. Thiagarajan et al. (2022) obtained sufficient diversity by using input space anchoring to sample a fully stochastic network. However, in the context of Bayesian neural networks (BNNs), it was recently shown that partial stochasticity can perform equally well with respect to fully stochastic BNNs at significantly less cost (Sharma et al., 2023). This suggests that in addition to the "amount" of diversity, the "effective" or functional diversity is also important for performance. However, in practice, it is difficult to control this balance, so existing methods default to heuristics that only promote diverse hypotheses. For example, DeepEnS uses different random seeds or shuffles the batch order when creating ensemble members, and Δ -UQ relies upon fully stochastic models. To this end, we propose three different anchoring strategies that only handle the difficulties of working with graph data and GNNs, but also induce different scales of the aforementioned balance. At a high-level, our strategies trade-off the amount of stochasticity (i.e., amount of diversity) and the semantic expressivity of the anchoring distribution to accomplish this.

Notations. Let $\mathcal{G} = (\mathbf{X}^0, \mathbf{A}, Y)$ be a graph with node features $\mathbf{X}^0 \in \mathbb{R}^{N \times d}$, adjacency matrix $\mathbf{A} \in [0, 1]^{N \times N}$ and labels Y , where N, d, q denote the number of nodes, feature dimension and number of classes, respectively. When performing graph classification, $Y \in \{0, 1\}^q$; for node

¹For example, channel wise concatenating two RGB images creates a 6 channel sample.